

1	Course Name:
	DATA SCIENCE ESSENTIAL
2	Course Code:
	IBB32403
3	Year and Semester Offered:
	Year 2 Semester 1
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	. This course is a study of the generalization of knowledge from data. The integrated skill set required are mathematics, statistics, machine learning, databases and a good understanding of the problem formulation to engineer effective solutions. This course will introduce the students to the data scientist job where they need to deal with various of data science practices, including data collection and integration, exploratory data analysis, predictive and descriptive modeling, evaluation and effective communication. Students will learn concept, techniques and tools to work with data.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Explain the basic principles of Data Science. 2. Prepare a clean data to be analyzed 3. Build models to produce prediction.
8	Content Outline:
	Topic
	1. Introduction to Data Science
	2. Basic Usage
	3. Exploratory Data Analysis
	4. Data preparation
	5. Building better processes
	6. Predictive Model Algorithms
	7. Model Construction and Evaluation
9	References:
	1. Principles of Data Science by Sinan Ozdemir, December 2016, ISBN 13 : 9781785887918 2. Analytics in a Big Data World: The Essential Guide to Data Science and its Applications by Bart Baesens, May 2014 (First Edition), ISBN13 : 978-1118892701

